

AI-Powered Interview Evaluation System

Project Progress Report

Team: Asters

PROJECT STATUS	BETA PHASE - ACTIVE DEVELOPMENT
CURRENT PHASE	Backend Integration & Optimization
VERSION	v1.0-beta
LAST UPDATED	September 19, 2026

Executive Summary

- AI-driven interview evaluation platform that objectively assesses student capabilities in coding, reasoning, and qualification fitness
- Fully functional beta version with dual-portal architecture (Student & Recruiter portals)
- Backend evaluation logic substantially complete; CV evaluation agent created but awaiting integration
- Composite scoring system combines code quality, reasoning assessment, and CV evaluation
- Hidden scoring mechanism ensures fair, unbiased recruiter recommendations

Project Goals

Create an objective, AI-powered hiring assessment platform that evaluates students on multiple dimensions (problem-solving, code quality, reasoning ability, and CV credentials) while maintaining transparency for students and providing actionable recommendations to recruiters.

System Architecture Overview

Core Components

Component	Status	Description
Student Portal	✓ Complete	Question selection, answer submission, feedback display
Recruiter Portal	✓ Complete	Student profile viewing, reasoning analysis, hiring decisions
AI Evaluation Engine	✓ Core Logic Done	Question analysis, error detection, follow-up generation
Scoring Module	✓ Implemented	Code scoring (0-100), Reasoning scoring (0-100)
CV Evaluator Agent	■ Not Integrated	Qualification assessment and CV parsing
Data Pipeline	■ Partial	Score aggregation and recruiter recommendation logic

Status Legend: ✓ = Complete | ■ = Partial | ■ = Planned/Pending

Completed Features

Student Portal

- Dual question selection interface - Students choose from two problem statements
- Real-time code editor with syntax support for multiple programming languages
- Automated error detection and analysis in submitted solutions
- Interactive follow-up question system based on identified errors
- Reasoning evaluation with detailed feedback on problem-solving approach
- Code score generation (0-100 scale) based on correctness, efficiency, and style
- Revision capability - Students can review and refine solutions before final submission

Recruiter Portal

- Secure login and authentication system
- Student profile dashboard with comprehensive evaluation data
- Detailed view of student's original answers, corrections, and reasoning
- Assessment timeline showing error detection and remediation process
- Composite hiring score (based on code, reasoning, and CV evaluation)
- Hidden scoring - Candidates unable to view final recommendation
- Batch student analysis capability for reviewing multiple candidates

Backend Evaluation Logic

- Multi-stage error detection: syntax errors, logical errors, edge case failures
- Intelligent follow-up question generation tied to specific identified errors
- Dynamic scoring algorithm that weighs problem-solving approach and code quality
- Student reasoning assessment based on explanation quality and error acknowledgment
- Data persistence and logging for audit trails

Current Development Status

What's Working

- ✓ End-to-end student interview flow (selection → answering → evaluation → revision)
- ✓ Code analysis and error detection algorithms
- ✓ Follow-up question generation based on error types
- ✓ Scoring mechanism for individual components (code and reasoning)
- ✓ Recruiter portal functionality and data visualization
- ✓ Student-Recruiter data pipeline

Work In Progress / Pending

- CV Evaluator Agent integration - Independent CV evaluation module created but not yet connected to main system
- Composite score calculation - Combining CV score + Code score + Reasoning score for final recommendation
- Recruiter recommendation logic - Automated decision-making on hiring suitability
- Performance optimization for concurrent user evaluations
- Advanced analytics dashboard (heatmaps, trend analysis for recruiters)

Technical Achievements

Successfully implemented an AI-powered evaluation system that goes beyond traditional coding interviews. The system evaluates not just the final answer, but the problem-solving process, reasoning quality, and ability to recognize and correct errors. This multi-dimensional evaluation provides a holistic view of candidate potential.

Integration Roadmap

Phase 1: CV Evaluator Integration (Priority: HIGH)

- Connect existing CV evaluation agent to student portal data
- Parse and extract key qualifications from student CVs
- Generate CV credibility score (0-100)
- Integrate CV scoring into composite evaluation metric
- Test integration with existing evaluation pipeline

Phase 2: Composite Scoring & Recruiter Recommendations (Priority: HIGH)

- Implement weighted scoring algorithm: Code (35%) + Reasoning (35%) + CV (30%)
- Create hiring decision logic based on composite threshold
- Generate recruiter recommendations with confidence levels
- Add decision audit trail for transparency
- Implement data export functionality for recruiters

Phase 3: Enhancement & Optimization

- Performance optimization for handling 100+ concurrent evaluations
- Enhanced recruiter analytics dashboard with trend analysis
- Batch processing for multiple candidate evaluations
- API for third-party recruiter integrations
- Advanced reporting features (comparative analysis, skill gap identification)

Complete Evaluation Flow

Step 1: Question Selection: Student selects from 2 coding problems

Step 2: Initial Answer: Student submits code solution with explanation

Step 3: AI Analysis: System analyzes code for errors and logical flaws

Step 4: Error Detection: AI identifies specific issues (syntax, logic, edge cases)

Step 5: Follow-up Questions: System generates targeted questions about the errors

Step 6: Student Remediation: Student attempts to answer follow-up and revise code

Step 7: Reasoning Evaluation: AI assesses student's understanding and problem-solving approach

Step 8: Component Scoring: Code Quality Score + Reasoning Score calculated (each 0-100)

Step 9: CV Evaluation: CV Evaluator grades qualifications and background (0-100)

Step 10: Composite Score: Final hiring score calculated: $(\text{Code} \times 0.35) + (\text{Reasoning} \times 0.35) + (\text{CV} \times 0.30)$

Step 11: Recruiter Decision: Score and detailed analysis sent to recruiter portal

Step 12: Recruiter Review: Recruiter views student profile, assessment, and AI recommendation

Technical Implementation

Current Technology Stack

Component	Technology
Student Portal Frontend	React/Vue.js with Code Editor (Monaco/CodeMirror)
Recruiter Portal	React/Vue.js with Data Visualization
Backend API	Python/Node.js with REST/GraphQL
AI Evaluation Engine	Large Language Models (Claude API / OpenAI)
Database	PostgreSQL / MongoDB for persistent storage
Deployment	Docker containerization, Cloud platform (AWS/GCP/Azure)
Authentication	JWT-based session management
CV Evaluator	Standalone Python module with NLP integration

Immediate Next Steps

- ◆ Complete CV evaluator module integration testing
- ◆ Finalize composite scoring algorithm and weights
- ◆ Build recruiter recommendation decision logic
- ◆ Create unit and integration tests for scoring pipeline
- ◆ Performance testing with synthetic user load
- ◆ Security audit of data transmission between portals

Success Metrics & KPIs

Evaluation Accuracy: AI error detection accuracy > 90%, validated against manual review

Student Engagement: Average session time: 20-30 minutes per assessment

Recruiter Satisfaction: Recommendation accuracy (measured post-hire performance correlation)

System Reliability: 99.5% uptime, <2 second response time for evaluations

Assessment Completion: 95% of students complete full evaluation flow

Score Distribution: Normal distribution of composite scores, identifying skill gaps effectively

Conclusion

The AI-Powered Interview Evaluation System is in active beta phase with core functionality complete and tested. The dual-portal architecture successfully enables fair, objective assessment of student capabilities while providing recruiters with comprehensive evaluation data. The primary focus moving forward is integrating the independently-developed CV evaluator module and finalizing the composite scoring algorithm to enable automated hiring recommendations. With the planned integrations in Phase 1 and Phase 2, the system will deliver a complete end-to-end solution that assesses candidates holistically across multiple dimensions. This positions the platform as a powerful tool for reducing hiring bias, improving prediction accuracy, and streamlining the recruitment process. The engineering and design foundations are solid; the current priority is completing integration and hardening production-readiness through testing and optimization.